Assessing the Impact of Data Cleaning on the Relationship Between Hospital Quality Ratings and Community Social Vulnerability

IS537 – Theory and Practice of Data Cleaning
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1. Project Summary

This project examines whether data cleaning decisions affect conclusions about the relationship between hospital quality ratings and community-level social vulnerability. The analysis integrates two datasets: the CMS Hospital General Information dataset (January 2026) and the CDC Social Vulnerability Index (SVI) dataset (2022). The focus of the project is to evaluate how data quality issues, particularly inconsistencies in geographic identifiers, impact dataset integration and fitness-for-use.

The central research question is:
Does cleaning and standardizing geographic identifiers improve integration quality and influence downstream analysis outcomes?

2. Dataset Description

The primary dataset is the CMS Hospital General Information dataset, which contains hospital-level attributes such as facility name, address, state, county, and overall hospital rating. The dataset includes 5,426 hospitals.

The secondary dataset is the CDC Social Vulnerability Index (SVI), which provides county-level measures of social vulnerability based on socioeconomic and demographic characteristics. This dataset contains approximately 3,144 counties and uses FIPS codes as standardized geographic identifiers.

These datasets originate from different federal agencies (CMS and CDC) and reflect distinct data collection processes, making them appropriate for integration analysis.

3. Original Data Quality Goals

The project initially aimed to address key data quality challenges, including:

- Consistency: Variations in county naming conventions (e.g., presence/absence of suffixes such as “County” or “Parish”)
- Validity: Typographical errors and inconsistent formatting in geographic fields
- Completeness: Missing hospital ratings
- Integration Quality: Absence of a shared geographic identifier (FIPS) in the CMS dataset

The primary goal was to standardize geographic fields and enable reliable integration between datasets.

4. Progress and Deliverables

Significant progress has been made across all stages of the project.

Data Profiling

Initial profiling identified the structure and quality of both datasets. The CMS dataset contains over 5,426 hospital records, while the SVI dataset includes over 3,144 counties. A notable finding is that approximately 47% of hospitals have missing overall ratings (“Not Available”), indicating a substantial completeness issue that may affect downstream analysis.

```
Hospital overall rating
Not Available    2560
3                 935
4                 765
2                 649
5                 288
1                 229
Name: count, dtype: int64
```

Figure 1: Not Available hospital ratings is 47%

Initial Integration Attempt

A naive join using county and state fields resulted in a match rate of approximately 93.3%, indicating inconsistencies in geographic naming and formatting across the two datasets.

```
Total hospitals: 5426
Matched: 5061
Naive Match rate: 0.9327312937707335
```

Figure 2: Initial Join rate is 93% before cleaning and FIPS integration

FIPS-Based Integration

To enable reliable integration, a U.S. Census county reference file was used to map county and state combinations to official FIPS codes. This step introduced a standardized geographic key for joining with the SVI dataset.

Using this approach, FIPS codes were successfully assigned to almost 100% of hospitals, representing a significant improvement in integration readiness.

```
Total hospitals: 5481
Hospitals mapped to county FIPS: 5476
New FIPS mapping rate: 0.9990877577084474
```

Figure 3: FIPS mapping of HGI.csv after cleaning is 99.9%

Data Cleaning and Standardization

To improve integration quality, multiple cleaning steps were implemented:

- Standardization of county names using uppercase formatting and removal of suffixes such as “County,” “Parish,” and “Borough”
- Removal of punctuation and normalization of spacing
- Correction of typographical errors (e.g., “Jeffrson” to “Jefferson”)
- Handling of abbreviations (e.g., “E. Baton Rouge”)
- Application of targeted fixes for specific edge cases (e.g., Washington, D.C., Alaska boroughs, and select counties with naming inconsistencies)

After joining with the SVI dataset using FIPS codes, the final join rate was approximately **98%**, with a small number of unmatched records remaining.

```
Final HGI-to-SVI join rate: 0.9799
```

Figure 4: Final Join rate is 98% after cleaning and FIPS integration

5. Evidence of Progress

Progress is supported by the following artifacts:

- Jupyter notebook documenting data loading, profiling, cleaning, and integration steps
- Intermediate datasets and final merged dataset
- Summary metrics capturing improvements in join rates
- Outputs identifying unmatched records after cleaning

These artifacts demonstrate measurable improvements in data quality and integration.

Github Link - https://github.com/kritikasingh62/IS537-Data-Cleaning-Project

6. Changes in Scope

No major changes in scope were made. However, the project placed greater emphasis on FIPS-based integration after identifying limitations in direct string-based joins. This shift improved both reproducibility and integration accuracy.

7. Challenges and Next Steps

Key challenges encountered include:

- Structural differences in geographic classifications (e.g., Alaska boroughs and U.S. territories)
- High proportion of missing hospital ratings
- Minor inconsistencies that cannot be fully resolved through rule-based cleaning

The next steps will focus on:

- Comparing results before and after FIPS-based integration (naive join vs cleaned join)
- Evaluating how data cleaning affects dataset coverage and usability
- Analyzing the remaining unmatched records after FIPS mapping to understand structural limitations in integration

References

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4. Use of Chat GPT to assist in code generation and format the structure of this document